Project – Building a Media Computing Application

Deliverables		Due Time
Proposal	Electronic submission	**11:59pm, 30 Mar (Mon, Week 06)**
	Hardcopy submission	In the lecture of Week 06
Final	Electronic submission	**11:59pm, 18 May (Mon, Week 12)**
	Hardcopy submission	In the lecture of Week 12
	Presentation	**6:00pm, 19 May (Tue, Week 12)**

INTRODUCTION

This project assignment is worth **30%** of the total assessment of this course, including **10%** for the proposal phase (Proposal) and **20%** for the final phase (Final).

This is a **group** assignment and each group consists of up to **THREE** members (could be **FOUR** members subject to the course coordinator's approval). If you hope to complete the assignment by yourself or with more members, you should first consult with course coordinator in written form.

Each group member must contribute to the assignment actively and equally with their contributions clearly stated in the Proposal and the Final Report, and the members will be awarded the same marks. Under certain circumstances, adjustment of marks may happen to group members at the discretion of the course coordinator.

TASK

You are required to identify, develop, and present an application of digital media computing (e.g., audio/image/video processing, computer vision, computer graphics, and animation). The applications can be of the following topics (but not limited to):

1. Special Visual/Audio Effects (e.g., re-sizing, de-hazing, de-blurring, morphing, and other enhancements)
2. Digital media understanding (e.g., object detection, recognition and tracking, action recognition, fake image detection, media forensics, and remote sensing)
3. Computer graphics and animation, VR, AR
4. 3D vision and robotics

Sample Applications

1. D. Glasner, et al., Super-Resolution from a Single Image, ICCV, 2009. http://www.wisdom.weizmann.ac.il/~vision/single_image_SR/files/single_image_SR.pdf
2. A. Criminisi, et al., Region Filling and Object Removal by Exemplar-based Image Inpainting, IEEE Trans. on Image Processing, 13(9): 1 – 13, Sep 2004. https://www.irisa.fr/vista/Papers/2004_ip_criminisi.pdf

This assignment should be completed in two phases: Proposal phase and Final phase. **It is highly encouraged to conduct thorough research and come up with a solid plan in the proposal phase.**

Proposal Phase

You are required to identify an application and complete a proposal for accomplishing the application. In the proposal, you need to articulate the following components, but not limited to:

1. Describing the application: background, significance, related work, and etc;
2. Describing technical method: key technical aspects of the solution to finish this project;
3. Describing the implementation plan: tasks and schedule; and
4. Providing a reflection on completing the proposal, such as how to form ideas and solutions and how to compose a good proposal.

Note that it is **NOT necessary** to follow the above headings in your proposal, but your proposal needs to be self-contained with such relevant content. You could imagine that the purpose of the proposal is to convince readers of your dream project. A good proposal will at least tell readers: What is your application about, such as developing an image super-resolution method (either an existing one or a new one)? Why is your application important? What is the relationship between your application and similar applications or method? What is your solution? What is the timetable or schedule of your implementation? ...... **Your solution must be based on an existing method (e.g., one of those listed under Sample Applications) of which no source codes have been released publicly.**

In the proposal, a section of **Reflection** is required to explain what resources you have consulted and what guidelines you have applied or/and what tools you have utilised when completing proposal (including both forming ideas and solutions and composing or writing the proposal) for improve your learning. You are strongly encouraged to obtain some guidance on proposal writing from a wide range of resources (e.g. books, articles, and websites on how to write a proposal).

The proposal file must be of **Adobe Acrobat Portable Document Format (*.pdf)** format. No other file format is accepted. The proposal must **NOT** exceed **TEN** pages including references, following the template provided. A hardcopy of the submitted proposal is required for submission. A template for ICME 2021 conference is available in Canvas.

Final Phase

You are required to complete a digital media computing application and deliver: 1) a final workable program (including all the source codes) or artwork (including all the source files and documents), 2) a short (around 3 to 5 minutes) introduction video of your project, 3) a readme or manual (if necessary), and 4) a final report and presentation slides.

The video must be compatible with the VLC media player. The report and manual files must be of **Adobe Acrobat Portable Document Format (*.pdf)**. No other file format is accepted. The final report must **NOT** exceed **TWELVE** pages including references, following the template provided. A hardcopy of the final report and the manual files is also required for submission. A template for ICME 2021 conference is available in Canvas.

The report should include a summary or abstract and at least provides: Introduction (e.g., What is your application about? Why is your application important?), Related Work (e.g., What is the relationship between your application and similar ones?), Methodology or Solution (e.g., What are the specific functions of your application? What are the solutions?), and Results (e.g., What are the datasets and What are the outcomes or results?). You are encouraged to follow the template of scientific articles listed in Sample Applications.

The manual is to guide a potential user on how to set up the working environment of your application and re-produce the results presented in your report.

MARKING SCHEME

Notes on Using AI

When using AI tools to complete some tasks (e.g., coding and report writing), you need to:

1. provide a record of using AI as a supplementary document in PDF format. For example, the input to the AI tool, the prompt used, and the output of the AI tool. It is not necessary to record every conversation with or every step using AI, however, the record of key progresses or changes using AI should be provided.
2. Provide a reflection of using AI to improve learning as described below (e.g., Point 4 in Proposal and Point 6 in Final).

Proposal (10 marks)

1. (2 marks) Application: creativity, challenge, and/or novelty of the application domain with justification.
   [2 marks] highly interesting and challenging application or task
   [1~1.5 marks] very interesting and challenging application or task
   [0 ~ 0.5 mark] simple application or task

2. (5 marks) Method: quality and feasibility of the solution (e.g. tasks).
   [4 ~ 5 marks] A technical method requiring efforts similar to a complicated published paper
   [2.5 ~ 3.5 marks] A technical method requiring reasonable efforts similar to a good published paper
   [0 ~ 2 marks] A technical method with little challenge

3. (1 mark) Plan: clarity of the individual tasks.
   [1 mark] clear timeline for specific tasks
   [0 ~ 0.5 mark] reasonable overview of the project

4. (2 marks) Reflection: reflection on how to write a good proposal, and reflection on using AI to improve learning in completing this proposal.
   [2 marks] High quality reflection
   [1 ~ 1.5 marks] Thoughtful reflection
   [0 ~ 0.5 mark] Meaningless reflection

Final (20 marks)

To pass this phase of the assignment, your application should at least be workable.

1. (8 marks) Method: Application and technical solution
   Assess the creativity, challenge, and quality of the application and solution
   [7.5 ~ 8 marks] A technical method requiring efforts similar to a very complicated published paper
   [6 ~ 7 marks] A technical method requiring efforts similar to a complicated published paper
   [4 ~ 5.5 marks] A technical method requiring reasonable efforts similar to a good published paper
   [0 ~ 3.5 marks] A technical method with little challenge

2. (5 marks) Results: Experimental results
   [4 ~ 5 marks] Comprehensive evaluation and investigation
   [2.5 ~ 3.5 marks] Sufficient benchmarking and discussions
   [0 ~ 2 marks] limited results with limited information on experimental settings

3. (1 mark) Documentation: Assignment Report, and a manual for computing programs
   You need to provide both a manual to document information which will assure other users to compile, install, and run your programs, or reproduce your artwork, and an assignment report document including background, system design, solution, results, discussions (if necessary), etc.
   [1 mark] high quality video presentation of the project
   [0 ~ 0.5 mark] a video presentation with limited efforts (e.g., simple recording of voice-over slides)

4. (1 mark) Introduction video of your project (around 3 minutes in MP4 format, less than 30MB).
   [1 mark] high quality video presentation of the project
   [0 ~ 0.5 mark] a video presentation with limited efforts (e.g., simple recording of voice-over slides)

5. (3 marks) Presentation (Refer to the Notes to Presentation in the Appendix)
   You need to deliver an oral presentation on your application in the lecture of the due week and the presentation will be marked in terms of clarity, understanding of the topic, and presentation skills. The presentation file must be included in the electronic submission.
   Presentation content (1 mark):
   [1.5 marks] high quality presentation content (e.g., professional typesetting, clear logic flow with high quality visual content,)
   [1 mark] good presentation content (e.g., clear logic flow with clear and relevant visual content)
   [0 ~ 0.5] poor presentation content (e.g., too much text or meaningless visual content)

   Presentation manner (1 mark):
   [1.5 marks] excellent presentation manner, such as language fluency, time, pace, and eye contact
   [1 mark] good presentation manner, such as time, pace, and eye contact
   [0 ~ 0.5] poor presentation manner, such as time, pace, and eye contact

6. (2 marks) Reflection: reflection on using AI to improve learning in completing the final stage of this project (e.g., method, coding, report writing, and presentation).
   [2 marks] High quality reflection
   [1 ~ 1.5 marks] Thoughtful reflection
   [0 ~ 0.5 mark] Meaningless reflection

SUBMISSION

- One submission is required from each group. Please **use your student ID (SID) and unikey, but NOT your full name,** in the documents such as reports and source codes.
- For electronic submissions, choose the correct submission under the section *Assignments*: ***Project Proposal*** and ***Project Final***.
- For the electronic submission of ***Project Final***, make the following two submissions:

  - Submit your project documents to ***Project Final [Document Only]***.
    Zip document files of your project (e.g. introduction video, readme file/manual, presentation slides, and report) and use your unikey as the name of the zip file.
  - Submit all your project materials to ***Project Final [All Materials]***.
    Zip all your project files (e.g. final output such as programs, source codes, introduction video, readme file/manual, presentation slides, and report) and use your unikey concatenated with "**_all**" as the name of the zip file. Make sure that the zip file less than **100MB**.
- There may be a penalty if 1) you do not include your SID, unikey, and the URL of the complete zip file (if applicable), in your documents, 2) your zip file cannot be accessed and unzipped successfully, or 3) your programs cannot be compiled and run successfully, or your artwork and introduction video cannot be played/ reproduced successfully.
- Refer to course materials and relevant policies on assessment issues, such as **late submission**, **special consideration**, and **plagiarism**.

RESOURCES

1. Computer Vision related Conferences sponsored by Computer Vision Foundation
   http://openaccess.thecvf.com/menu.py
2. ACM International Conference on Multimedia (ACM MM)
   https://dblp.uni-trier.de/db/conf/mm/index
3. IEEE International Conference on Multimedia & Expo (ICME)
   https://dblp.uni-trier.de/db/conf/icmcs/index.html
4. ACM SIGGRAPH
   https://dblp.uni-trier.de/db/conf/siggraph/index
   https://dblp.uni-trier.de/db/conf/siggrapha/index
   http://kesen.realtimerendering.com/
5. Arts and Interactions
   OpenTK (http://www.opentk.com/)
   NUI gropup (http://nuigroup.com)
   ARToolkit (http://www.hitl.washington.edu/artoolkit)
   Processing (http://processing.org)
   openFrameworks (http://www.openframeworks.cc)

Appendix – Notes to Project Presentation

For project presentation, each group has maximally **8 minutes** to present the assignment using the presentation material submitted. The presentation will generally be starting at 6:00pm from Group 1 in the weekly lecture room. You should be familiar with the presentation facility in the room. Detailed presentation schedule will be available in due course.

Either the computer in the lecture room or your personal laptop is allowed for presentation. If you use the computer in the lecture room, your presentation materials should be compatible with the settings in the lecture room and **be copied to the computer before 6:00pm on the presentation day**. If you use your personal laptop, you must 1) comply with safety regulations of the University of Sydney, and 2) have successful trialed with the presentation facilities available in the lecture room.

Any delay may lead to penalty in marking, or even losing presentation opportunity. Should you have any difficulty, please feel free to contact Lecturer.

This presentation session which is worth 3 marks of the 20 marks for the whole assignment, will be assessed in terms of Presentation content and skills (e.g. logical flow, understanding, visual/audio aids, time, pace, and eye contact, and clear explanation of results (e.g., quantitative and/or qualitative results, and demonstration)). Please refer to Marking Scheme for more details.